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ORIGINAL PAPER

Application of bioremediation in restoring the polluted Cikakembang river

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Abstract

Rapid industrial expansion and population growth in Indonesia have led to severe water pollution in the Cikakembang River, a tributary of the Citarum River. This study evaluated the potential of microbial bioremediation as a cost-effective and sustainable approach to improve water quality. The experiment evaluated the effects of bacterial type, bacterial ratio, and aeration across twelve treatment combinations. Key water quality parameters such as dissolved oxygen (DO), chemical oxygen demand (COD), ammonia nitrogen (NH3–N), nitrate (NO_3^-), total nitrogen (TN), and total phosphate (TP) were observed over a 9-day period, with measurements taken every three days. The results showed that bioremediation, particularly when combined with aeration, significantly improved water quality, achieving notable reductions in COD and nitrogen compounds. Significant improvements were achieved under aerated conditions, with average reductions of 82.45% COD, 78.94% NH3–N, 71.83% NO_3^-, 33.49% TN, and 95.62% TP. Combinations B2 and B3, using the first bacterial set and aeration at ratios of 1:50,000 and 1:100,000, showed the highest pollutant removal efficiency. These findings demonstrate the strong potential of microbial bioremediation with aeration for reducing organic and nutrient pollutants in river systems similar to the Cikakembang River.

Keywords Cikakembang river · Environmental biotechnology · Microbial bioremediation · Water quality improvement · Water pollution control

Introduction

The rapid urban growth and industrial expansion that have occurred in Indonesia over recent decades has driven significant economic progress. However, these developments have also placed severe pressure on natural resources, especially water bodies (Fitriana et al. 2023; Tyagi & Kumar 2020). A particularly striking example is the Citarum River, which has been designated as one of the ten most polluted rivers worldwide (Blacksmith Institute 2008). The river is severely contaminated, with an estimated daily discharge of 340,000 tons of wastewater from approximately 2000 textile factories (Permatasari et al. 2022). The presence of various domestic activities within the watershed has been identified as a contributing factor to the severity of pollution, leading to the introduction of additional contaminants and wastes into the river (Idris et al. 2019).

According to data from the Ministry of Environment and Forestry, 54% of the Citarum River's water is heavily polluted while only 3% meets the water quality standard (Indonesian Ministry of Environment and Forestry 2018). Three cascade dams—Saguling, Cirata, and Jatiluhur—which are fed by the Citarum River, have also been polluted, affecting the life of the aquatic ecosystem (Wardhani & Sugiarti 2021). Notwithstanding the severely degraded water quality of the Citarum River, it is designated as a strategic river because it still supplies approximately 80% of the drinking water needs of Jakarta as the capital city of the country (Marselina et al. 2025). The adverse impacts of Citarum River pollution are reflected in the increased incidence of

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dermatological diseases such as itching, skin irritation, and diarrhea among nearby residents (Salami et al. 2025).

Significant measures have been undertaken by the Indonesian Government to address water quality degradation, one of which is the enactment of Law Number 32 of 2009. This law emphasizes environmental management through the regulation of hazardous and toxic waste handling (Indonesian Government, 2009). Another strategy to improve the Citarum River's water quality was the Citarum Bestari Movement, a multiparty action plan (RAM-IP) supported by Government Regulation No. 78/2015 (Idris et al. 2019). Unfortunately, the water quality of the Citarum River remains poor and fails to meet water quality standards. This is essentially the result of the ineffective execution of the pollution control programs (Djuwita et al. 2021). Thus, a different alternative is required to improve the current condition of the Citarum River.

Bioremediation has gained global attention for its low energy requirements, simplicity, and ability to avoid secondary pollution (Saeed et al. 2022). In China, microbial bioremediation has been successfully applied to several polluted rivers, such as the Chengnan and Xuxi Rivers, significantly improving water quality (Gao et al. 2018; Wu & Xie 2012; Yudianto & Xie 2011), demonstrating its potential for application in the Citarum River. A detailed implementation process is outlined in Yudianto and Xie (2011), which includes sample collection, laboratory analysis to identify primary pollutants, selection of appropriate bacterial cultures and concentrations, field preparation, bacterial injection into the water body, and regular monitoring. A study conducted by Wu & Xie (2012) reported that the use of microbial technology in the Xuxi River in Wuxi City led to significant reductions in chemical oxygen demand (COD), total phosphate (TP), and ammonia nitrogen (NH3–N) by 43%, 56%, and 58%, respectively, while the average dissolved oxygen (DO) levels increased from 0.42 mg/L to 2.4 mg/L (Wu & Xie 2012). Similarly, in 2018, the application of microbial technology in the Chengnan River in Nanjing City resulted in a DO increase to 5 mg/L, with COD, TP, and NH3–N decreasing by 38%, 15%, and 20%, respectively (Gao et al. 2018). Furthermore, the utilization of specific bacterial strains has demonstrated efficacy in the degradation of pesticides classified as Persistent Organic Pollutants (POPs) (Ayi 2024). These compounds frequently elude detection and enter water bodies such as rivers and lakes through various means, including direct discharge from agricultural fields or infiltration into groundwater. Pesticide-contaminated water can subsequently lead to the accumulation of these pollutants in fish, posing potential health risks to humans through consumption. In-situ bioremediation employing *Bacillus* bacteria has naturally degraded these POPs along with other organic pollutants (Ayi 2024). Consequently, a substantial decline in pollutant concentrations contributes to the restoration of ecological balance, thereby minimizing the health risk to humans and promoting the enhancement of aquatic biodiversity (Kuppan et al. 2024).

Considering the success of previous studies, this research applies microbial bioremediation technology to restore the water quality of the Citarum River. This approach is selected due to Indonesia's tropical climate and relatively constant temperatures throughout the year, which increase the likelihood of bacterial survival in water bodies for extended periods. Moreover, the limited availability of land in urban areas restricts the construction of conventional treatment facilities. Therefore, this study was designed as a pilot project to provide preliminary evidence supporting the future implementation of in-situ bioremediation. The objectives of this study are (1) to evaluate the effectiveness of bioremediation in reducing organic pollutants and (2) to determine the optimal treatment method for maximum efficiency based on the laboratory experiments.

Materials and methods

Study location

The Cikakembang River is one of the tributaries with significant potential to contribute to pollution in the upper watershed of the Citarum River (Kent et al. 2025). The river is located in the Majalaya District, which is dominated by the textile industry, with around 174 factories currently in operation (Kent et al. 2024). The absence of a centralized wastewater treatment plant (WWTP) in the Majalaya District has resulted in each textile factory managing its wastewater independently. Moreover, the lack of comprehensive regulatory oversight raises concerns about the adequacy and consistency of these individual treatment practices. Concurrently, domestic wastewater from neighboring residential areas is discharged directly into the Cikakembang River without undergoing prior treatment. As illustrated in Fig. 1, the study area encompasses the distribution of textile factories and residential dwellings along the Cikakembang River. As illustrated in the figure, the silver-hued rooftops symbolize textile factories, whereas the brick-red rooftops denote residential housing.

As shown in Fig. 2, the sampling point was selected at the downstream section of the Cikakembang River, just before entering the main body of Citarum River, with coordinates of 7°01′28.6"S, 107°43′51.0"E. In addition, tributaries are basically preferred for bioremediation implementation due to their manageable physical characteristics, which facilitate more effective intervention and monitoring.

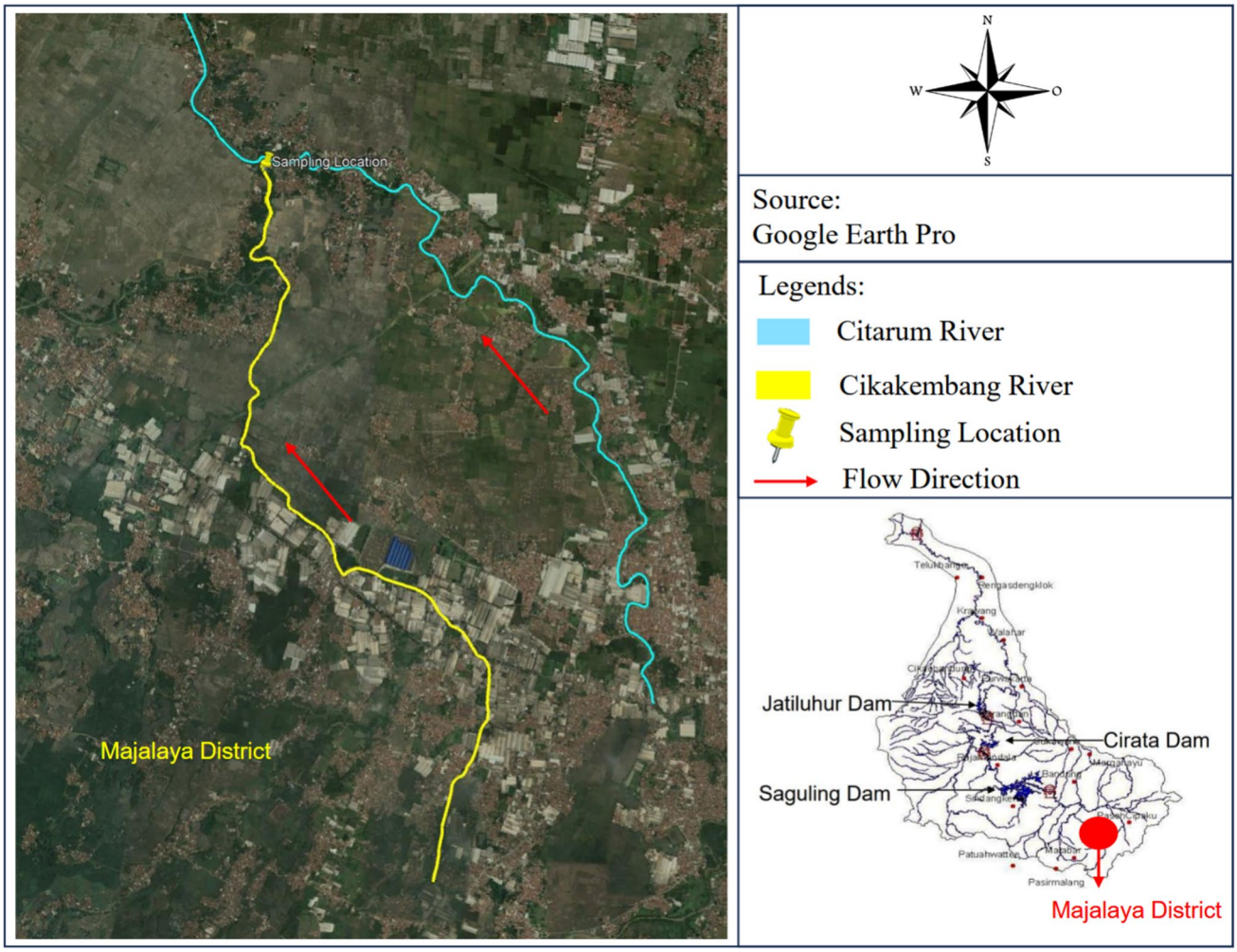

Fig. 1 Study location and spread of textile factories and residentials along the Cikakembang River (the index Map was adapted from Boer et al. 2015)

Effluent contributions to the Cikakembang river

The textile industry wastewater is characterized by elevated concentrations of COD and biological oxygen demand (BOD) (Azanaw et al. 2022; Tebeje et al. 2022). Untreated effluents have been reported to exhibit COD values ranging from 150 to 30,000 mg/L and BOD values ranging from 80 to 6,000 mg/L (Yaseen & Scholz 2019). In contrast, domestic wastewater typically contains elevated concentrations of nutrient parameters, particularly nitrogen and phosphorus compounds. (Rout et al. 2021). According to Widyarani et al. (2022), domestic wastewater in Indonesia typically contains NH_3–N concentrations ranging from 0.7 to 20 mg/L and TP concentrations of approximately 24 mg/L. A study by Fitriana et al. (2023) analyzed COD and ammonia data from the Cikakembang River between 2013 and 2021, revealing that COD levels exceeded the Indonesian Class II water quality standard of 25 mg/L on 17 occasions between 2013 and 2017. While there was an observed improvement in the COD levels between 2017 and 2021, these levels did not reach the established standard. A slight discrepancy was observed in the parameter of ammonia, with only six instances recorded in 2013 and 2014, indicating concentrations that exceeded the 0.5 mg/L threshold. However, subsequent to 2015 and extending through 2021, the ammonia concentration remained within acceptable limits (Fitriana et al. 2023). A separate study applied the QUAL2Kw model to simulate COD and NH3–N levels in the Cikakembang River (Fitriana et al. 2024). According to the findings of the aforementioned study, the estimated loads of pollutants ranged from 500 to 2,600 kg per day for COD and from 16 to 86 kg per day for NH3–N.

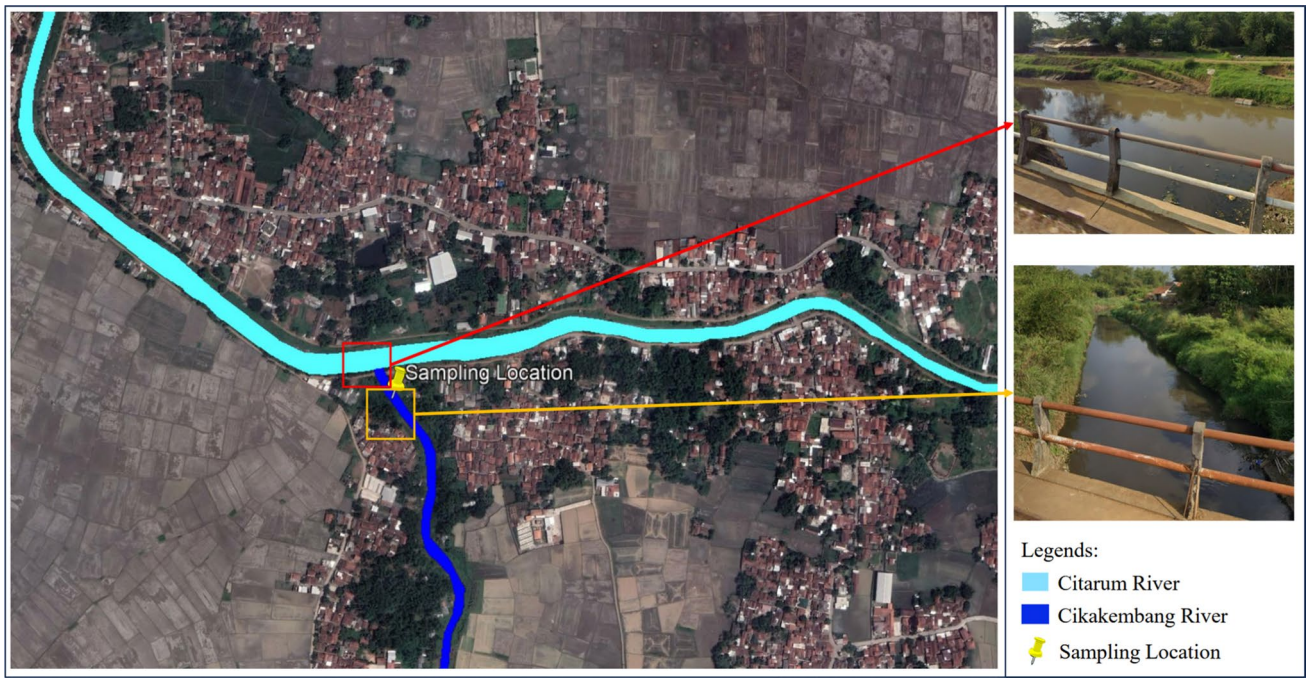

Fig. 2 Sampling location

Methodology

A field investigation was conducted to assess the parameters of organic pollutants at the study location. A total of 200 L of water samples were collected during the morning hours, specifically between 8:30 and 9:00 a.m. This specific time period was deliberately chosen to align with the discharge of wastewater from the textile industry, thereby ensuring sufficient mixing of pollutants within the river's flow. In this study, the bioremediation process was conducted ex-situ, whereas the water samples were transferred to the laboratory for further experimentation. This method was selected to ensure better control over treatment scenarios, including variations in bacterial types, ratios, and aeration conditions. This approach was also intended to minimize ecological risks. Two bacterial sets were tested: the first consisted of *Bacillus subtilis*, *Bacillus licheniformis*, *Bacillus giganteum*, and *Bacillus gelatinus*; the second included *Bacillus subtilis*, *Bacillus licheniformis*, and an alkaline protease enzyme. Each set was applied at three concentration ratios. Six samples were aerated, resulting in twelve experimental combinations, as summarized in Table 1. Each test used 8 L of water, and laboratory analyses were performed every three days, with the final measurement on Day 9. The nine-day monitoring period was chosen to capture the initial phase of microbial adaptation and pollutant removal efficiency under controlled conditions. The experimental workflow is illustrated in Fig. 3.

Table 1 Combinations of experimental test

Combination	Notes			
	Type of bacteria	Aerator	Ratio	Total bacteria (gr)
B1	1	Yes	1/10,000	0.80
B2	1	Yes	1/50,000	0.16
B3	1	Yes	1/100,000	0.08
B4	2	Yes	1/10,000	0.80
B5	2	Yes	1/50,000	0.16
B6	2	Yes	1/100,000	0.08
B7	1	No	1/10,000	0.80
B8	1	No	1/50,000	0.16
B9	1	No	1/100,000	0.08
B10	2	No	1/10,000	0.80
B11	2	No	1/50,000	0.16
B12	2	No	1/100,000	0.08

Treatment methods

Wastewater treatment methods can generally be categorized as physical, chemical, or biological (Hanafi & Sapawe 2020). Biological treatment utilizes microorganisms such as bacteria and fungi to degrade organic compounds (Kanaujiya et al. 2019), while physical and chemical methods rely on physical forces (e.g., membrane filtration, adsorption) and chemical reactions (e.g., coagulation, oxidation),

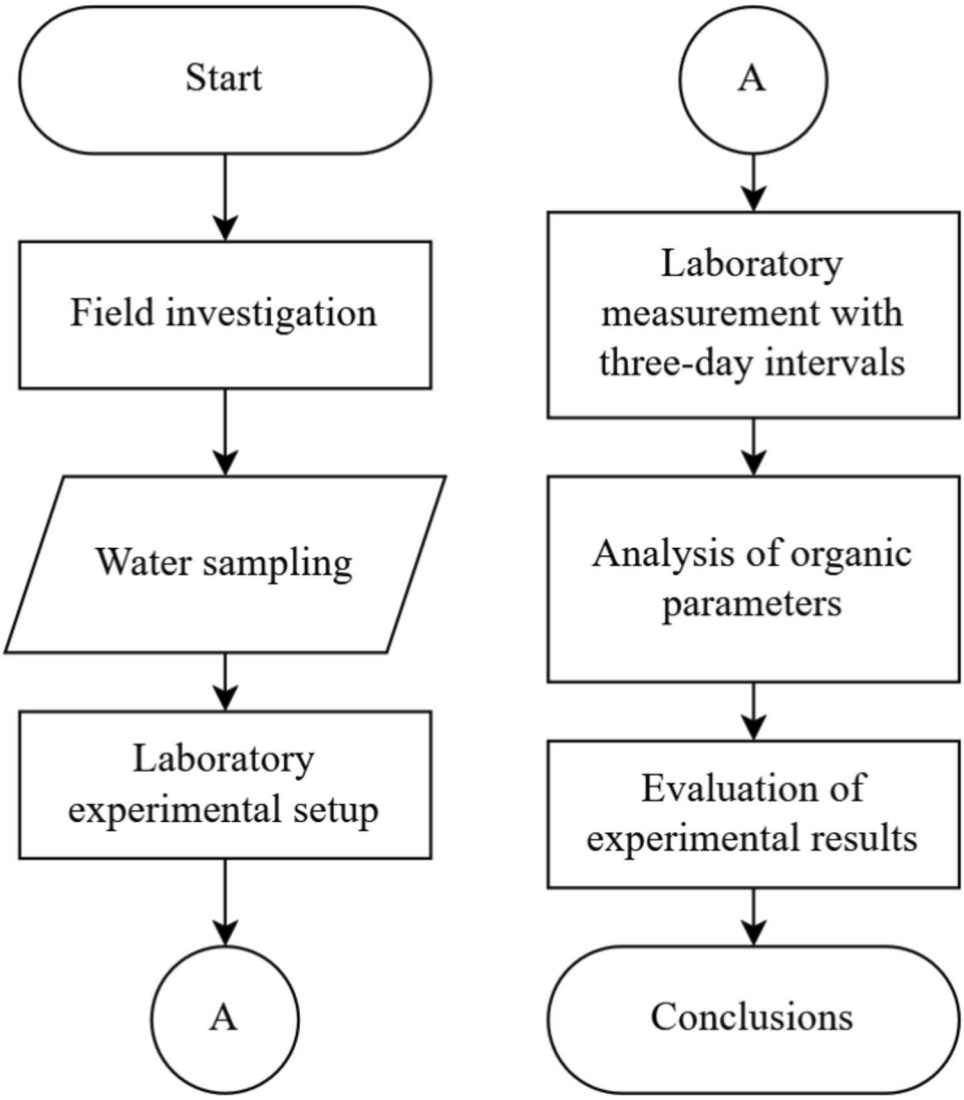

Fig. 3 The research flowchart

respectively (Liu et al. 2025; Włodarczyk-Makuła et al. 2023). Several studies have explored hybrid systems combining these approaches to improve treatment efficiency (Farabegoli et al. 2004; Tuan et al. 2023). Farabegoli et al. (2004) reported that biological treatment using activated sludge achieved pollutant removal efficiencies comparable to those of the Fenton–adsorption process, but at much lower operational costs. Consequently, biological processes are often preferred for their cost-effectiveness and sustainability. Recent reviews also highlight their suitability for textile wastewater (Kato & Kansha 2024), although large spatial requirements—particularly for systems such as constructed wetlands—remain a limitation in densely populated urban areas (Ma et al. 2024).

Bioremediation, a form of biological treatment, is widely recognized as a cost-effective method that requires minimal space. It employs microorganisms—primarily bacteria or fungi—to degrade, neutralize, or remove pollutants from contaminated environments. As described earlier, this study used four bacterial species—*B. subtilis*, *B. licheniformis*, *B. giganteum*, and *B. gelatinus*— and an enzyme, *alkaline protease*. *B. subtilis* has been shown to remove heavy metals such as lead and chromium through metabolic processes that promote the formation of organic–carbonate complexes, thereby immobilizing metal ions (Ahn et al. 2023). It also degrades complex organic compounds via enzyme secretion (Zhang et al. 2020). Similarly, *B. licheniformis* contributes to the biodegradation of industrial waste through enzymes such as proteases and amylases, and its resilience under extreme conditions enhances its bioremediation potential (Gopinath et al. 2017; Sarwan & Bose 2022). Both *B. subtilis* and *B. licheniformis* are also recognized as safe probiotics for human use (Lefevre et al. 2017; Nithya et al. 2012).

B. giganteum was included for its ability to degrade lignin and complex aromatic compounds (Wróbel et al. 2023), while *B. gelatinus* was chosen for its effectiveness in

decomposing gelatinous and proteinaceous materials (Sarwan & Bose 2022). Both species are non-pathogenic and categorized under Biosafety Level 1 (BSL-1), indicating minimal risk to human health and the environment (Munjal et al. 2016; U.S. Environmental Protection Agency 2001). The *Alkaline proteases enzyme* complements these bacterial activities by efficiently hydrolyzing protein-rich waste and demonstrating strong stability under extreme conditions, making it suitable for industrial and wastewater treatment applications (Jafari et al. 2023; Kotb et al. 2023; Meena et al. 2024; Saeed et al. 2022). Like the bacterial species used in this study, alkaline protease is also considered safe for human applications (Varia et al. 2019).

Laboratory test methods

This study primarily evaluated six water quality parameters, including DO, COD, NH3–N, TN, NO_3^-, and TP. The laboratory testing procedures for these parameters are outlined in Table 2. The DO concentration was measured using a membrane electrode, with the method's accuracy set to ± 0.1 mg/L and its precision set to ± 0.05 mg/L (Baird et al. 2017). The COD parameter was determined via the dichromate reflux method using potassium dichromate ($K_2Cr_2O_7$), sulfuric acid (H_2SO_4), and mercuric sulfate ($HgSO_4$) solutions, followed by spectrophotometric analysis; duplicate samples were required to differ by less than 5% (Baird et al. 2017). The NO_3^- concentration was analyzed using the ion chromatography system, in which 0.6068 g of sodium nitrate ($NaNO_3$) were dissolved in reagent water (Pfaff & Hautman 1997). The phenate method, a widely employed technique in the field for testing NH3–N, involves the reaction of ammonia, hypochlorite, and phenol in the presence of a sodium nitroprusside catalyst, resulting in the formation of an intense blue compound (Baird et al. 2017). Subsequently, the sample's degree of light absorption will be measured at a wavelength of 640 nm by means of a spectrophotometer. A more intense blue coloration is indicative of a higher concentration of NH3–N. The TN parameter was analyzed using the chromotropic acid method, which has been demonstrated to be applicable for concentrations ranging from 1 to 50 mg/L (Baird et al. 2017). The TP was analyzed according to the Indonesian National Standard (SNI) 6989–31:2021, using persulfate oxidation to convert all phosphorus species to orthophosphate, which was then reacted with ammonium molybdate and potassium antimonyl tartrate for colorimetric detection.

Table 2 Water quality parameter assessment methods

Parameter	Unit	Method
DO	mg/L	Membrane electrode method
COD	mg/L	Digestion reactor method
NO_3^-	mg/L	EPA method 300.1 ion chromatography
NH3–N	mg/L	Phenate method
TN	mg/L	Chromotropic acid method
TP	mg/L	SNI 6989–31:2021 (part 31)

Water quality parameters

Dissolved oxygen (DO)

DO is defined as the oxygen present in the atmosphere or produced through the process of photosynthesis that dissolves in water, where it supports aquatic life and participates in various organic reactions, reflecting the river's self-purification capacity (Ali et al. 2022; Nugraha et al. 2020). DO is a critical indicator of water quality in streams, rivers, and lakes, with higher DO concentrations generally signifying better water quality (American Water Works Association, 2017). DO levels exhibit variation in accordance with water pressure, temperature, and salinity and they are typically measured in milligrams per liter (mg/L) or as a percentage of saturation (DO%).

Chemical oxygen demand (COD)

The COD is a method of measuring the amount of oxygen required to oxidize organic material present in a water sample (Rekrak et al. 2020). A high level of COD concentration is indicative of severe water contamination (Wu et al. 2022), which can result in oxygen depletion or hypoxia within aquatic systems due to microbial degradation processes (Lv et al. 2024). Hypoxia has the potential to adversely impact fishery production, as dissolved oxygen (DO) concentrations become inadequate to sustain aquatic life. Another adverse effect is the occurrence of eutrophication in water bodies exhibiting elevated COD concentrations (Kashem et al. 2023). Furthermore, it has been established that water with elevated COD levels is unsuitable for consumption and domestic use (Wiyantoko et al. 2015).

Nitrate (NO_3^-)

NO_3^- represents the end product of nitrogen oxidation in natural waters and is also the most soluble form of nitrogen compounds. Its presence in surface waters is often associated with the use of fertilizers, representing the end product of nitrogen oxidation in natural waters (Bijay-Singh & Craswell, 2021).

Ammonia nitrogen (NH3–N)

NH3–N is composed of unionized ammonia and ammonium, with their proportions influenced by pH and temperature.

In most natural waters, NH3–N predominantly exists as ammonium, but as pH and temperature rise, the proportion of unionized ammonia increases (Lin et al. 2019). Elevated ammonia levels can lead to eutrophication, where abundant nutrients promote rapid algae growth. Ammonia can also be converted into nitrite (NO_2^-) and subsequently into NO_3^-, which are crucial indicators of water quality.

Total phosphate (TP)

Phosphorus, a leading cause of eutrophication, is present in water in various forms, including orthophosphate, condensed phosphate, and organic phosphorus compounds (Wang et al. 2023). A minimal quantity of phosphorus has the potential to induce substantial contamination of water bodies, thereby impeding the water's prospective applications.

Total nitrogen (TN)

TN refers to the accumulation of NO_3^-, NO_2^-, NH3–N, and other organic nitrogen forms. It has the same effects as other nitrogen compounds in affecting an aquatic environment, such as eutrophication (Nieder & Benbi 2022).

Indonesian river water quality standard

The quality standards for river water in Indonesia are delineated in Government Regulation No. 22 of 2021, which pertains to the implementation of environmental protection and management measures. The water quality standard is classified into four classes, one of which is class II, designated for water recreation, fisheries, livestock, and irrigation activities. The water quality standard is delineated in Table 3. According to the findings of the study conducted by Fitriana et al. (2024) the Citarum River is expected to comply with the Class II Indonesian River Water Quality Standard. Consequently, the established standards serve as benchmarks for evaluating the water quality parameters.

Table 3 Class II Indonesian river water quality standard (Indonesian Government, 2021)

Parameter	Concentration
DO (mg/L)	> 4.00
COD (mg/L)	< 25.00
NO_3^- (mg/L)	< 10.00
NH3–N (mg/L)	< 0.20
TN (mg/L)	< 15.00
TP (mg/L)	< 0.20

Calculation of reduction percentage (%)

The reduction percentage (RP) in water quality concentration can be computed using various principles. One of the principles calculates BIAS percentage between the two data. The resulting values determine whether the concentration of water quality parameters has increased or decreased. In this study, higher values indicate better results. The mathematical formula used to calculate these differences is provided in Eq. 1.

$$RP = 100 \cdot \left(\frac{C_0 - C_z}{C_0} \right) \tag{1}$$

where,

RP: Reduction percentage concentration of each water quality parameter (%),

C_0: Concentration of a water quality parameter on Day 0 (mg/L),

C_z: Concentration of a water quality parameter on Day 9 (mg/L).

Results and discussion

The water sample was collected on July 18, 2024. A preliminary visual inspection of the sample indicated normal turbidity, consistent with typical river conditions, as shown in Fig. 4a. Despite the apparent clarity, laboratory analyses presented in Table 4 revealed that none of the measured parameters met the applicable water quality standards. The most significant deviation was observed in the COD parameter, with a measured concentration of 180.2 mg/L—well above the standard limit of 25 mg/L. A similarly substantial deviation was noted in ammonia nitrogen, with a percent bias from the standard reaching 97.73%. Concentrations of NO_3^-, TN, and TP also exceeded standard values, measured at 63 mg/L, 18.85 mg/L, and 2.00 mg/L, respectively. Furthermore, the DO concentration was found to be significantly below the standard threshold of 4 mg/L, with a recorded value of 1.67 mg/L.

As demonstrated in Fig. 4b, where all combinations underwent treatment, the water exhibited a conspicuous enhancement in clarity in comparison to its initial state. Laboratory test results also confirmed an improvement in water quality. Among the 12 treated combinations, variations were observed across all parameters, particularly in COD, NH3–N, and DO. As illustrated in Fig. 5, a comprehensive evaluation of the alterations in water quality parameters for each combination is provided.

Following the initial water quality assessment, bacteria were introduced into 12 samples according to the specified

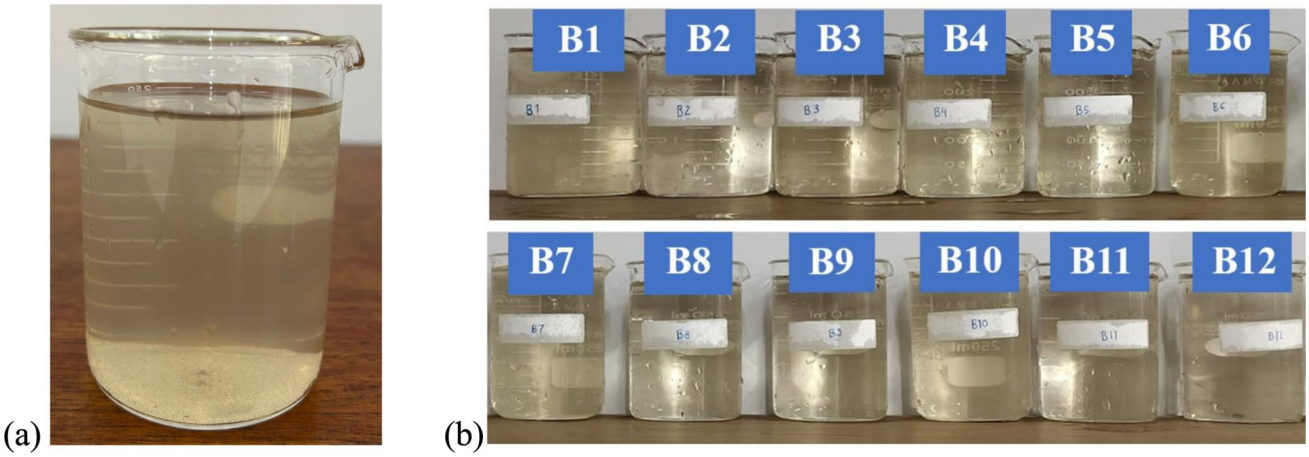

Fig. 4 The Cikakembang river water sample **a** by Day 0 and color of each test combination **b** by Day 9

Table 4 Laboratory test results for water quality parameters on day 0

Parameter	Unit	Day-0	Class II river water quality standard
DO	mg/L	1.67	> 4.00
COD	mg/L	180.20	< 25.00
NO_3^-	mg/L	63.00	< 10.00
NH3–N	mg/L	8.83	< 0.20
TN	mg/L	18.85	< 15.00
TP	mg/L	2.00	< 0.20

ratios and bacterial sets. By the third day, notable variations were observed in the water quality parameters across all combinations. DO levels exhibited a slight increase, with concentrations ranging from Combinations B1 to B6 showing higher levels due to the use of aerators. Concentrations of COD remained constant, while NO_3^- levels exhibited a substantial decrease across all combinations. However, the utilization of aerators appeared to delay the degradation of NO_3^-, as evidenced by the observations in Combinations B1 to B6. Concentrations of NH3–N exhibited a slight increase, while TN and TP concentrations demonstrated a general decrease. In contrast, Combination B6 exhibited an opposite trend, characterized by a decline in NH3–N levels and an increase in TN levels.

By the sixth day, a marked improvement in DO levels was observed across all combinations. Combinations B1 to B6 showed a modest increase in DO, while Combinations B7 to B12 demonstrated a more pronounced rise. Variations in COD were also noted, with several combinations showing lower values than those recorded on Day 3, indicating effective bioremediation and active degradation of organic pollutants. As time progressed, a decline in NO_3^- concentrations was observed, following a similar pattern to that of COD. NH3–N levels varied depending on aeration: combinations equipped with aerators (B1 to B6) showed a decrease, whereas non-aerated combinations (B7 to B12) exhibited an increase. TN levels, which had declined on Day 3, showed an upward trend by Day 6. TP concentrations fluctuated, with Combinations B1, B4, and B7 showing lower values compared to Day 3, while the remaining combinations displayed either higher or relatively stable levels.

By the ninth day, a notable improvement in overall water quality was observed, marked by a substantial increase in DO levels. A comprehensive review of the extant literature reveals a strong correlation between the presence of aerators and the substantial increase in DO levels. COD levels decreased markedly from 180.2 mg/L on Day 0 to around 20 mg/L. Regarding nitrogen compounds, bacteria notably reduced NH3–N, with Combination B3 achieving the lowest level of 0.1 mg/L. Although NO_3^- and TN levels were reduced, the decrease was not as substantial as COD and NH3–N. TP concentrations fell significantly across all combinations, with the lowest values recorded in Combinations B5 and B6 at 0 mg/L. Table 5 presents the percentage reduction values for each water quality parameter to comprehensively illustrate the final water quality improvements for each combination.

Based on the overall results shown in Fig. 5, distinct trendlines for each water quality parameter were observed. The test results were categorized into four groups: (1) samples treated with the first type of bacteria set and equipped with an aerator (B1 to B3), (2) samples treated with the second type of bacteria set and equipped with an aerator (B4 to B6), (3) samples treated with the first type of bacteria set, but without an aerator (B7 to B9), and (4) samples treated with the second type of bacteria set, but without an aerator (B10 to B12). This classification was based on similar patterns observed on the ninth day, where each set of three combinations exhibited comparable characteristics. The trendlines for these four groups are presented in Fig. 6.

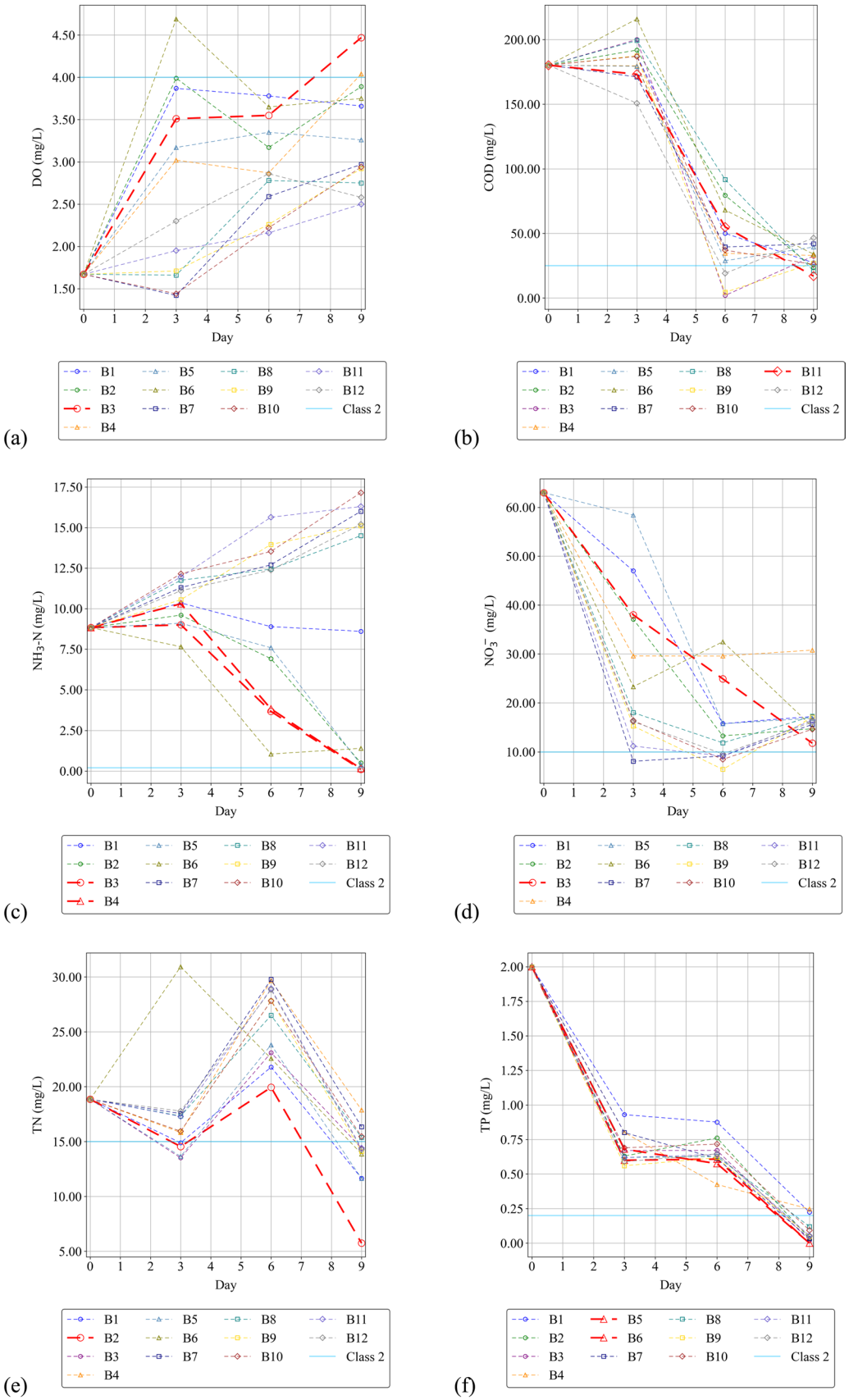

Fig. 5 Assessment Result of **a** DO, **b** COD, **c** NH3–N, **d** NO_3^-, **e** TN, and **f** TP over a 9-days period

Table 5 Reduction percentage for all combinations

Combination	Parameter				
	COD	NO_3^-	NH3–N	TN	TP
B1	84.85%	72.70%	2.55%	38.13%	88.80%
B2	86.96%	76.59%	94.33%	69.51%	98.28%
B3	81.96%	81.29%	98.87%	23.59%	98.98%
B4	81.69%	51.13%	98.30%	5.09%	87.68%
B5	78.08%	73.27%	95.47%	38.23%	100.00%
B6	81.13%	76.00%	84.14%	26.37%	100.00%
B7	76.69%	75.19%	−81.30%	13.27%	98.38%
B8	88.35%	72.62%	−64.31%	18.41%	94.00%
B9	85.02%	72.87%	−71.10%	25.18%	97.55%
B10	85.57%	76.75%	−94.33%	17.94%	95.33%
B11	90.57%	73.89%	−84.70%	23.59%	97.28%
B12	74.20%	74.43%	−72.24%	24.25%	97.60%

In addition to visual or trendline analysis, the reduction in pollutant concentrations was quantitatively assessed, as shown in Table 5. Positive percentage values indicate a reduction in pollutant concentration, whereas negative values indicate an increase. The concentrations of COD and nitrogen compounds were significantly reduced, with average values of COD 82.45%, NH3–N 78.94%, NO_3^- 71.83%, TN 33.49%, and TP 95.62% in aerated combinations. However, NH3–N levels in the non-aerated combinations showed a substantial increase. The variation in NH3–N concentrations under non-aerated conditions suggests that oxygen limitation may have hindered nitrification, leading to incomplete ammonia oxidation. In contrast, aerated combinations exhibited a consistent decline in NH3–N levels, emphasizing the importance of aerobic conditions in facilitating nitrogen transformation processes. According to Yudianto & Xie (2011), the microbial decomposition of nitrogenous compounds often leads to elevated NH3–N concentrations, which can be effectively reduced in subsequent treatment stages.

Discussions

There are two parameters that should be considered from the trendline analysis results: the gradient, which indicates the overall increase or decrease over the 9-day period, and the intercept, which represents the initial value. Among these two parameters, the gradient is more critical than the intercept, as it plays a greater role in visualizing the reduction in water quality parameter concentrations. The trendlines shown in Fig. 6 highlight the significant effect of aerator use in the bioremediation experiments. Combinations that utilized an aerator—represented in the figure as TL B1–B3 and TL B4–B6—exhibited greater gradients, particularly for DO, NH3–N, NO_3^-, and TN parameters. In contrast, for the TP parameter, the use of an aerator did not appear to influence the rate of concentration reduction. Additionally, TL B1–B3 generally demonstrated better performance compared to TL B4–B6, which is consistent with the reduction percentages previously presented in Table 5.

In terms of reduction percentage calculations, Combinations B2 and B3 achieved the highest average values, at 85.13% and 76.94%, respectively. With regard to adherence to the established standard, Combinations B2 and B3 demonstrated success in meeting three and five of the six parameters that were subjected to evaluation, respectively. This finding underscores the hypothesis that Combination B3 demonstrates the greatest potential for application in the domain of water quality treatment, particularly in relation to uses associated with water recreation and fisheries. Among all parameters assessed in Combination B3, only COD exceeded the regulatory threshold; however, the deviation was minor, with a measured concentration of 32.5 mg/L, slightly above the permissible limit of 25 mg/L. The extension of the monitoring period could result in a decline in COD levels. Although this study demonstrated substantial reductions in multiple pollutant parameters within a nine-day period, the limited monitoring duration remains a constraint. The nine-day timeframe was intentionally chosen to allow for a rapid evaluation of the initial response and effectiveness of the bacterial treatment under controlled conditions. However, longer monitoring periods (e.g., 30–90 days) are recommended for future studies to better capture long-term dynamics, stability, and potential declines in COD and other pollutants.

As this study assessed water quality parameters only during specific times, precise magnitude of temporal fluctuations in these organic parameters, driven by the application of bacteria, could not be captured. Implementing a real-time water quality monitoring system would enhance the reliability and effectiveness of future bioremediation efforts. Several studies have effectively deployed water quality monitoring systems utilizing sensor-based in riverine, lacustrine, and coastal environments (Huan et al. 2020; Koronides et al. 2024; Ramadhan 2020; Ramírez et al. 2019). These studies have shown that sensor-based systems deliver reliable measurements, often comparable with conventional laboratory methods. It is important to note, however, that sensors vary in their parameter detection capabilities (Yaroshenko et al. 2020); therefore, sensor selection should be carefully tailored to the monitoring objectives.

There are three considerations that must be addressed before implementing in-situ bioremediation. First, the potential environmental impacts of introducing external microbial species must be carefully evaluated. As emphasized by Yudianto and Xie (2011), a rigorous screening process is essential to ensure that only indigenous bacterial strains are

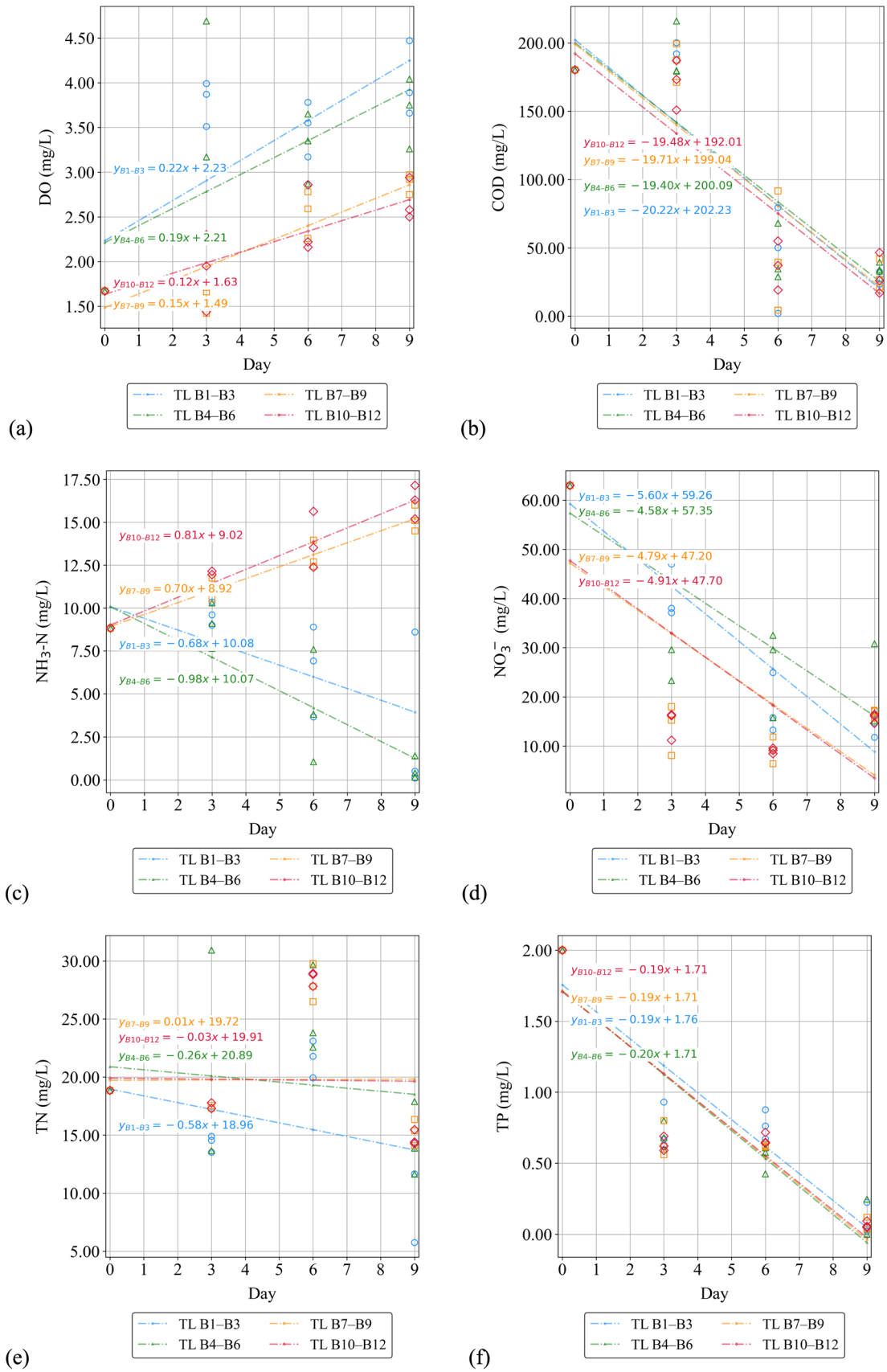

Fig. 6 Trendline of each group for **a** DO, **b** COD, **c** NH3–N, **d** NO_3^-, **e** TN, and **f** TP parameters

employed. In this context, future studies should focus on evaluating and characterizing the native microbial communities in the Cikakembang River and Citarum River before initiating in-situ bioremediation efforts. Second, the development of a numerical model is also essential to support the preliminary treatment process. Such a model can simulate pollutant loadings, evaluate various treatment scenarios, and estimate the optimal bacterial dosage required. This modeling framework minimizes trial-and-error during field implementation, thereby improving decision-making and enhancing the overall efficiency of bioremediation strategies. To ensure the reliability of model outputs, comprehensive supporting data from both rivers should be incorporated. Thirdly, although the effects may be minimal, the implementation of in-situ bioremediation needs to be synchronized with seasonal conditions, particularly since Indonesia only experiences two distinct seasons, namely dry and rainy. These seasonal variations can influence bacterial growth, required bacterial dosage, and the natural self-purification capacity of the river.

Conclusion

As one of the major contributors to the pollution of the Citarum River, the Cikakembang River represents a critical target for remediation using alternative methods such as bioremediation to reduce organic pollutant concentrations. This study was conducted through a nine-day laboratory experiment with sampling at three-day intervals. Laboratory results demonstrated substantial reductions in all key parameters, with average decreases of 82.92% for COD, 73.06% for NO_3^-, 26.96% for TN, and 96.15% for TP. In contrast, NH3–N concentrations exhibited inconsistent trends across treatment combinations, showing marked improvements only in those incorporating aeration (Combinations B1–B6).

The most effective treatment combinations were identified for each pollutant parameter. Combination B2 achieved the highest reductions in COD (86.95%) and TN (69.51%), whereas Combination B3 provided superior performance in reducing NO_3^- (81.29%), NH3–N (98.87%), and TP (98.98%). Combinations B7–B12 were excluded due to their limited effectiveness in NH3–N removal. These findings indicate that the selection of an appropriate treatment configuration should be guided by site-specific conditions to ensure optimal pollutant removal efficiency prior to field-scale implementation.

Overall, all water quality parameters were effectively reduced using the first bacterial consortium. Experimental results further demonstrated that aeration significantly enhanced pollutant removal efficiency compared with non-aerated conditions. Among the tested dosages, bacterial ratios of 1:50,000 and 1:100,000 yielded the most substantial improvements in water quality. For future applications, the integration of sensor-based monitoring and numerical modeling is strongly recommended to optimize in-situ bioremediation performance, as these tools can markedly enhance process efficiency, operational control, and scalability. These findings provide a scientific basis for integrating microbial bioremediation into river management frameworks and may inform future environmental policies aimed at achieving sustainable water quality improvement in Indonesia.

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Data availability Not applicable.

Declarations

Conflict of interest The authors declare that they have no financial or non-financial interests that are relevant to the content of this article. All authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest, or non-financial interest in the subject matter or materials discussed in this manuscript. Additionally, no author is an editorial board member, editor, or guest editor in the journal where this manuscript is submitted, ensuring that there are no conflicts regarding the editorial review or publication decisions.

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